



1. Company and Product Identification

- 1.1 Identification – Product Name: **Avista 177**
- 1.2 Other means of identification: Organic Alkaline mixture
Synonym: Mixture, none
Recommended Use of the Chemical: Reverse osmosis membrane cleaner.
1.3 and Restrictions On Use: Use only as directed on the label.
- 1.4 Name, Address, And Telephone Number of the Manufacturer, Or Other Responsible Party: **AVISTA TECHNOLOGIES**
140 Bosstick Street
San Marcos, CA 92069
(760) 744-0536
Competent Person email address: regulatory@avistatech.com
- 1.5 24 Hour Emergency No.: 1-800-424-9300 (United States)
1-703-527-3887 (International Collect)



DRINKING WATER TREATMENT ADDITIVES CLASSIFIED BY NSF INTERNATIONAL TO ANSI/NSF 60 AS STANDARD DRINKING WATER TREATMENT CHEMICAL FOR USE OFF-LINE IN REVERSE OSMOSIS SYSTEMS

2. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW: This product is a clear, pale to slightly white corrosive liquid. This product may irritate or injure contaminated tissue, depending on concentration and duration of contact. Depending on the duration of contact, over-exposures can severely irritate or cause injuries to the eyes. This product is neither reactive nor flammable. Thermal decomposition of this product produces irritating vapors and toxic gases (e.g. carbon monoxide and carbon dioxide). Emergency responders must wear personal protective equipment (and have appropriate fire-extinguishing protection) suitable for the situation to which they are responding.

Physical Hazards Summary None

Potential Health Hazards Summary Skin Corrosion/Irritation Cat 1B
Eye damage/Irritation Category 2A
Specific target organ toxicity - single exposure Category 3 Respiratory system
Acute Toxicity, Oral Category 4
Acute Toxicity, Dermal Category 5

Potential Ecological Effects Summary None

2.1 Classification of Product

U.S. OSHA classification Skin Corrosion/Irritation Cat 1B
Eye damage/Irritation Category 2A
Specific target organ toxicity - single exposure Category 3 Respiratory system

Classification as per EC 1272/2008 (CLP/GHS)	Acute Toxicity, Oral Category 4
	Acute Toxicity, Dermal Category 5
	Skin Corrosion/Irritation Cat 1B
	Skin Corrosion/Irritation Cat 1B
	Eye damage/Irritation Category 2A
WHMIS classification	Specific target organ toxicity - single exposure Category 3 Respiratory system
	Acute Toxicity, Oral Category 4
	Acute Toxicity, Dermal Category 5
	Skin Corrosion/Irritation Cat 1B
	Eye damage/Irritation Category 2A
Hazardous Materials Information System (HMIS) Rating	Specific target organ toxicity - single exposure Category 3 Respiratory system
	Acute Toxicity, Oral Category 4
	Acute Toxicity, Dermal Category 5

Health	2
Flammability	0
Physical Hazard	0
Protective Equipment	C

2.2 Label Elements OSHA/GHS

General Warnings	P101	If medical advice is needed, have product container or label at hand.
	P102	
	P103	Keep out of reach of children.
	P403	Read label before use
	P233	Store in a well-ventilated place. Keep container tightly closed
Signal Word	DANGER	
Hazard statements	H314	Causes severe skin burns and eye damage
	H318	Causes serious eye damage
	H315	Causes skin irritation.
	H335	May cause respiratory irritation.
	H302	Harmful if swallowed
Precautionary statements	H313	May be harmful to skin
	P260	Do not breathe dust/fume/gas/mist/vapours/spray.
	P264	Wash thoroughly after handling.
	P280	Wear protective gloves/protective clothing
	P301+P330+	
	P331	IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
	P303+P361+P353	IF ON SKIN (or hair): Remove/Take off Immediately all contaminated clothing. Rinse SKIN with water/shower. Wash contaminated clothing before reuse.
	P363	IF INHALED: Remove victim to fresh air and Keep at rest in a position comfortable for breathing.
	P304+P340	IF exposed or if you feel unwell: P310 Immediately call a POISON CENTER or doctor/physician.
	P309+P310	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
	P305+	
	P351+P338	
Hazard pictograms	P405	Store locked up.
	P501	Dispose of contents/container according to local requirements



2.3	Unclassified Hazards	None
2.4	Ingredients with unknown acute toxicity	None

3. COMPOSITION and INFORMATION ON INGREDIENTS

Chemical name CAS # EINECS #	% w/w	US OSHA	GHS/EU CLP	WHMIS
Sodium Metasilicate 6834-92-0 215-687-4	30-40	Skin Corrosion/Irritation Cat 2 Eye damage/Irritation Category 2A H315 H318	Skin Corrosion/Irritation Cat 2 Eye damage/Irritation Category 2A H315 H318	Skin Corrosion/Irritation Cat 2 Eye damage/Irritation Category 2A H315 H318
Emulsifier 1 Proprietary Proprietary	5-10	Skin irritation (Category 2), Eye irritation (Category 2A), Specific target organ toxicity - single exposure (Category 3), Respiratory system, H315 H319 H335	Skin irritation (Category 2), Eye irritation (Category 2A), Specific target organ toxicity - single exposure (Category 3), Respiratory system, H315 H319 H335	Skin irritation (Category 2), Eye irritation (Category 2A), Specific target organ toxicity - single exposure (Category 3), Respiratory system, H315 H319 H335
Sulfite Salts Proprietary Proprietary	1-5	Acute Toxicity, Oral (Category 4) Acute Toxicity, Dermal (Category 5) Serious Eye Irritant (Category 2A) H302 H313 H319	Acute Toxicity, Oral (Category 4) Acute Toxicity, Dermal (Category 5) Serious Eye Irritant (Category 2A) H302 H313 H319	Acute Toxicity, Oral (Category 4) Acute Toxicity, Dermal (Category 5) Serious Eye Irritant (Category 2A) H302 H313 H319
Surfactant Proprietary Proprietary	1-5	Serious Eye Irritation Category 2; Aquatic Toxicity, chronic, Category 3 H319; H412	Serious Eye Irritation Category 2; Aquatic Toxicity, chronic, Category 3 H319; H412	Serious Eye Irritation Category 2; Aquatic Toxicity, chronic, Category 3 H319; H412
Emulsifier 2 Proprietary Proprietary	1-5	Not hazardous per OSHA	Not classified hazardous	Not classified hazardous
Acrylic Polymer Proprietary Proprietary	1-5	Not hazardous per OSHA	Not classified hazardous	Not classified hazardous
Polyacrylate Proprietary Proprietary	1-5	Not hazardous per OSHA	Not classified hazardous	Not classified hazardous
Water or other chemicals do not contribute to any additional hazards of this product	Balance			
PRODUCT CLASSIFICATION	100	Skin Corrosion/Irritation Cat 1B Eye damage/Irritation Eye irritation (Category 2A), Specific target organ toxicity - single exposure (Category 3), Respiratory system Acute Toxicity, Oral (Category 4) Acute Toxicity, Dermal (Category 5)		

NE = Not Established. C = Ceiling Limit. See Section 16 for Definitions of Terms Used.

The specific chemical identity and percentage are withheld as a trade secret.

4. FIRST-AID MEASURES

4.1 Description of Necessary Measures

Skin exposure: If this product contaminates the skin, immediately begin decontamination with running water. Remove exposed or contaminated clothing, taking care not to contaminate eyes. Victim should seek immediate medical attention if any adverse exposure symptoms develop.

Eye exposure: If this product enters the eyes, open victim's eyes while under gently running water. Use sufficient force to open eyelids. Have victim "roll" eyes. Minimum flushing is for 15 minutes. Victim must seek medical attention.

Inhalation: If dust of this product are inhaled, remove victim to fresh air. If necessary, use artificial respiration to support vital functions. Remove or cover gross contamination to avoid exposure to rescuers.

Ingestion: If this product is swallowed, CALL PHYSICIAN OR POISON CONTROL CENTER FOR MOST CURRENT INFORMATION. DO NOT INDUCE VOMITING. Have victim rinse mouth with water, if conscious. Never induce vomiting or give a diluent (e.g., water) to someone who is unconscious, having convulsions, or unable to swallow. If contaminated individual is convulsing, maintain an open airway and obtain immediate medical attention.

4.2 Most Important Symptoms/Effects: Immediate: Inhalation exposure may cause coughing or sneezing. Symptoms of skin and eye contact may include redness and irritation. Ingestion may cause stomach pains, cramps, and gastritis.

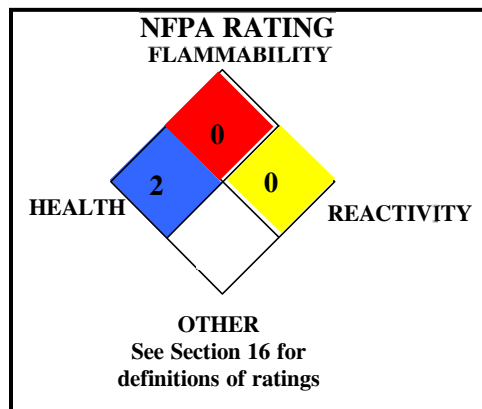
Delayed: Prolonged or repeated skin overexposure to this product may cause dermatitis (dry, red skin). Symptoms may include tingling, redness, and visible injury.

4.3 Indication of Immediate Medical Attention And Special Treatment Needed, If Necessary: **TARGET ORGANS:** Acute: Skin, eyes. Chronic: Skin.

Victims of chemical exposure must be taken for medical attention if any adverse effects occur. Rescuers should be taken for medical attention if necessary. Take a copy of label and MSDS to physician or health professional with victim.

5. FIRE-FIGHTING MEASURES

Flammable properties Non-flammable aqueous solution



Flash Point °C: Not applicable.

Autoignition Temperature °C: Not applicable.

Flammable Limits (in air by volume, %):

Upper: Not applicable.

Lower: Not applicable.

5.1 Suitable and Unsuitable Extinguishing Media:

This material will not contribute to the intensity of a fire. Use extinguishing material suitable to the surrounding fire.

Water spray	YES	Carbon dioxide	YES
Foam	YES	Dry chemical	YES
Halon	YES	Other	YES

5.2 Specific Hazards Arising from Chemical:

When involved in a fire, this material may decompose and produce irritating fumes and toxic gases (e.g., carbon monoxide, carbon dioxide, and nitrogen oxides).

Explosion Sensitivity to Mechanical Impact: Not applicable.

Explosion Sensitivity to Static Discharge: Not applicable.

5.3 Special Protective Equipment and Precautions for Fire-Fighters:

Incipient fire responders should wear eye protection. Structural firefighters must wear Self-Contained Breathing Apparatus and full protective equipment. Move containers from fire area if it can be done without risk to personnel. If possible, prevent runoff water from entering storm drains, bodies of water, or other environmentally sensitive areas.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal Precautions

Uncontrolled releases should be responded to by trained personnel using pre-planned procedures. Proper protective equipment should be used. In case of a spill, clear the affected area and protect people.

Protective equipment

For small releases (< 20 kg), clean up spilled liquid wearing gloves, goggles, face shield, and suitable body protection. The minimum Personal Protective Equipment recommended for response to non-incident releases (more than 20 kg) should be Level C: triple-gloves (neoprene gloves and nitrile gloves over latex gloves), chemical resistant suit and boots, hard hat, and full-face respirator with HEPA filter.

Emergency procedures

Monitoring must indicate that exposure levels are below those provided in Section 8 (Exposure Controls-Personal Protection) and that oxygen levels are above 19.5% before anyone is permitted in the area without Self-Contained Breathing Apparatus.

6.2 Methods and Materials for Containment and Cleaning Up

Moisten to suppress dust. Shovel up solids into plastic container for recovery/disposal. Soak up liquids with caustic-resistant media. Neutralize residue with sodium bicarbonate or other neutralizing agent for strong caustics. Decontaminate the area thoroughly. Test area with litmus paper to ensure neutralization. Place all spill residues in a suitable plastic container. Dispose of in accordance with applicable U.S. Federal, State, or local procedures, or appropriate local standards (see Section 13, Disposal Considerations).

7. HANDLING and STORAGE

7.1 Precautions for Safe Handling

All employees who handle this material should be trained to handle it safely. Open containers carefully on a stable surface. Empty containers may contain residual liquid; therefore, empty containers should be handled with care.

As with all chemicals, avoid getting this product ON YOU or IN YOU. Wash thoroughly after handling this product. Do not eat or drink while handling this material. Avoid generating dust or mist of this product. Remove contaminated clothing immediately.

During equipment maintenance follow practices indicated in Section 6 (Accidental Release Measures) to decontaminate equipment or clean-up small spills. Make certain that application equipment is locked and tagged-out safely if necessary. Collect all rinsates and dispose of according to applicable U.S. Federal, State, or local procedures or appropriate local standards.

7.2 Conditions For Safe Storage

Store containers in a cool, dry location, away from direct sunlight, sources of intense heat, or where freezing is possible. Store away from incompatible materials. Material should be stored in secondary containers, or in a diked area, as appropriate. Storage and use areas should be covered with impervious materials. Keep container tightly closed when not in use. If appropriate, post warning signs in storage and use areas. Inspect all incoming containers before storage, to ensure containers are properly labeled and not damaged.

Incompatibilities Strong acids.

8. EXPOSURE CONTROLS - PERSONAL PROTECTION

8.1 Control Parameters

CHEMICAL NAME	CAS #	% w/w	EXPOSURE LIMITS IN AIR				
			ACGIH-TLVs		OSHA-PELs		OTHER
			TWA	STEL	TWA	STEL	

			mg/m ³	mg/m ³	mg/m ³	mg/m ³	mg/m ³	mg/m ³
Sodium Metasilicate	6834-92-0	30	NE	NE	NE	NE	NE	NE
Emulsifier 1	Proprietary	10	NE	NE	NE	NE	NE	NE
Sulfite Salts anhydrous	Proprietary	5	5 A4 (Not Classifiable as a Human Carcinogen)	NE	NE	NE	NE	NIOSH REL: TWA = 5
Surfactant	Proprietary	5	NE	NE	NE	NE	NE	NE
Emulsifier 2	Proprietary	2	NE	NE	NE	NE	NE	NE
Acrylic Polymer	Proprietary	1.2	NE	NE	NE	NE	NE	NE
Polyacrylate	Proprietary	1	NE	NE	NE	NE	NE	NE
Water and other components which are present in less than 1 percent concentration (0.1% concentration for potential carcinogens, reproductive toxins, respiratory tract sensitizers and mutagens).		Balance	None of the other components contribute significant additional hazards at the concentration present in this product. All pertinent hazard information has been provided in this document, per the requirements of the Federal Occupational Safety and Health Administration Standard (29 CFR 1910.1200), U.S. State equivalent Standards and Canadian Workplace Hazardous Materials Identification System Standards (CPR 4).					

8.2 Appropriate Engineering Controls.

Use with adequate ventilation to ensure exposure levels are maintained below the limits provided in this Section or as low as reasonably practical. Ensure eyewash/safety shower stations are available near areas where this product is used.

8.3 Personal Protective Equipment

Respiratory protection:

None needed under normal conditions of use. Use NIOSH approved respirators if ventilation is inadequate to control mists or vapor. If respiratory protection is needed, use only protection authorized in the U.S. Federal OSHA Standard (29 CFR 1910.134), applicable U.S. State regulations, or the applicable local standards. Oxygen levels below 19.5% are considered IDLH by OSHA. In such atmospheres, use of a full-face piece pressure/demand SCBA or a full-face piece, supplied air respirator with auxiliary self-contained air supply is required under OSHA's Respiratory Protection Standard (1910.134-1998).

Eye protection: Use approved safety goggles or safety glasses, as described in OSHA 29 CFR 1910.133. Splash goggles with a face shield may be needed if splash hazards exist.

Hand protection: Wear chemical impervious gloves (e.g., Solvex™, Neoprene).

Body protection: If needed, use body protection appropriate for task (e.g., Tyvek suit, rubber apron) to protect from splashes and sprays.

9. PHYSICAL and CHEMICAL PROPERTIES

Appearance	This product is a clear, pale to amber		
Odor	None	Odor Threshold	N/A
Melting Point °C	NE	pH @ 2% solution	11.5 – 12.5
Initial Boiling Point °C	>100	Boiling Point Range °C	N/A
Flammability	Non-flammable	Evaporation Rate (water = 1)	Similar to water
Vapor Density (air = 1)	Similar to water	Vapor Pressure mm Hg @ 20°C:	Similar to water
Solubility (in water)	Completely miscible	Relative density (water = 1)	1.2 – 1.35
Viscosity	Similar to water	Oil-Water Partition Coefficient	N/A
Decomposition Temperature	NE		
How To Detect This Substance (Warning Properties):	Litmus paper will turn blue in contact with this solution.		

10. STABILITY and REACTIVITY

10.1	Reactivity	Not considered reactive.
10.2	Chemical Stability	Stable
10.3	Possibility of hazardous reactions	Hazardous polymerization will not occur.
10.4	Conditions to avoid	Avoid mixing with incompatible materials.

10.5	Incompatible Materials	Strong acids. It may react with metals to generate pressure.
10.6	Hazardous Decomposition Products	Thermal decomposition of this product may generate carbon monoxide and carbon dioxide.

11. TOXICOLOGICAL INFORMATION

Toxicity data for hazardous ingredients	Oral LD ₅₀ mg/kg	Dermal LD ₅₀ mg/kg	Inhalation LD ₅₀ mg/kg
Sodium Metasilicate	LD ₅₀ (Oral, rat) 1153	N/A	N/A
		Skin irritation (24 hr) severe	
Emulsifier 1	N/A	N/A	N/A
Sulfite Salts	LD ₅₀ (Oral, rat) = 1540 mg/kg	N/A	N/A
	Prolonged or repeated exposure may cause allergic reactions in certain sensitive individuals. Can cause pulmonary sensitization resulting in hyperactive airway dysfunction and pulmonary allergy accompanied by fatigue, malaise and aching. Significant symptoms of exposure can persist for more than two years and can be activated by a variety of nonspecific environmental stimuli such as automobile exhaust, perfumes and passive smoking.		
Surfactant	LD ₅₀ (Oral-Rat) > 2000 16,800 mg/kg	LD ₅₀ (Dermal-Rat) > 2000	N/A
	Skin/eye irritation (rabbit): non-irritant		
Emulsifier 2	N/A	N/A	N/A
	Sex Chromosome Loss and Nondisjunction (Oral-Drosophila melanogaster) 11 pph Standard Draize Test (Skin-rabbit) > 300 mg/kg		
Acrylic Polymer	N/A	N/A	N/A
Polyacrylate	N/A	N/A	N/A

12. ECOLOGICAL INFORMATION

ALL WORK PRACTICES MUST BE AIMED AT ELIMINATING ENVIRONMENTAL CONTAMINATION.

12.1	Ecotoxicity	LC ₅₀ , mg/L	EC ₅₀ , mg/L
	Sodium Metasilicate		
	Aquatic	Water Solubility = 59.2% (20°C); 84% (100°C) Biological Oxygen Demand (BOD): 40%, 5 days; 60%, 10-20 days. Food Chain Concentration Potential: Very Low Experimental Log P = -1.64 Persistence: Can ferment on standing. Biodegrades quite rapidly. It is dangerous to aquatic life in high concentrations. Lowers pH in water but does not dissociate to any great extent.	NE
	Terrestrial	NE	NE
	Emulsifier 1	NE	NE
	Sulfite Salts	LC50 - Gambusia affinis (Mosquito fish) - 660 mg/l - 96 h	
	Surfactant	NE	NE
		60.6 mg/L Fathead minnow (Pimephales promelas).	NE
		NE	NE
		NE, known to be toxic to aquatic organisms in high concentration	
	2- Emulsifier 2	N/A	N/A
	Acrylic Polymer	NA	N/A
	Polyacrylate	N/A	N/a
12.2	Persistence and Degradability	The components of this product decompose in soil and water.	
12.3	Bioaccumulative Potential	The components of this product are not expected to bioaccumulate.	

12.4	Mobility in Soil	When spilled onto soil, this product will infiltrate downward, the rate being greater with lower concentration because of reduced viscosity. During transport through the soil, this product will dissolve some of the soil material, in particular, carbonate-based materials.
12.5	Other Adverse Ecological Effects	This product may be harmful to aquatic life <u>if large volumes</u> of it are released into an aquatic environment.

13. DISPOSAL CONSIDERATIONS

Preparing Wastes of this Product for Disposal	Waste disposal must be in accordance with appropriate U.S. Federal, State, and local regulations or with local regulations. This product, if unaltered by the handling, may be disposed of by treatment at a permitted facility or as advised by your local waste regulatory authority.
Disposal of Contaminated Packaging	Cleaned containers can be recycled or disposed of as non-contaminated waste, if authorized by your local authorities. Dispose of containers as required by local regulations.
U.S. EPA Waste Number	D002. If this product is spilled or becomes a waste unchanged.

14. TRANSPORT INFORMATION

THIS MATERIAL IS HAZARDOUS AS DEFINED BY 49 CFR 172.101 BY THE U.S. DEPARTMENT OF TRANSPORTATION.

14.1	UN Number	UN3266
14.2	UN Proper Shipping Name	Corrosive Liquid, Basic, Inorganic, N.O.S., (Sodium Metasilicate)
14.3	Transport Hazard Class(es)	8 (Corrosive)
	Transport label(s) required	Corrosive Class 8
14.4	Packing Group	II
14.5	Marine Pollutant	Not applicable
	NA Emergency Response Guide Number (2018)	154
14.6	Transport in Bulk (Annex II of MARPOL 73/78 and IBC Code)	Not applicable
14.7	Special Transport Precautions	Not applicable
	National Motor Freight Classification	#70

International Air Transport Association

14.8	UN Number	UN3266
	UN Proper Shipping Name	Corrosive Liquid, Basic, Inorganic, N.O.S., (Sodium Metasilicate)
	Transport Hazard Class(es)	8 (Corrosive)
	Transport label(s) required	Corrosive Class 8
	Packing Group	II
	Packaging Instructions	850/851/852

International Maritime Organization

14.9	UN Number	UN3266
	UN Proper Shipping Name	Corrosive Liquid, Basic, Inorganic, N.O.S., (Sodium Metasilicate)
	Transport Hazard Class(es)	8 (Corrosive)
	Transport label(s) required	Corrosive Class 8
	Packing Group	II
	Marine Pollutant	Not applicable
	NA Emergency Response Guide Number (2018)	154
	Transport in Bulk (Annex II of MARPOL 73/78 and IBC Code)	Not applicable

15. SAFETY, HEALTH and ENVIRONMENTAL REGULATIONS SPECIFIC FOR THE PRODUCT

PROGRAM	Emulsifier 1	Sulfite Salts	Surfactant	Sodium Metasilicate	Propel- SCO-75	Acrylic Polymer	Polyacrylat e
US EPA PROGRAMS							
Clean Air Act Hazardous Air Pollutants	NO	NO	NO	NO	NO	NO	NO
Safe Drinking Water Act	NO	NO	NO	NO	NO	NO	NO
RCRA F, K, P, U or D-lists	NO	NO	NO	NO	NO	NO	NO
SARA 302 RQ	NO	NO	NO	NO	NO	NO	NO
SARA 302 TPQ	NO	NO	NO	NO	NO	NO	NO
SARA 313 LISTED	NO	NO	NO	NO	NO	NO	NO
SARA CHEMICAL CATEGORIES							
SARA 311/312 ACUTE	YES	YES	YES	YES	NO	NO	NO
SARA 311/312 CHRONIC	YES	YES	YES	YES	NO	NO	NO
SARA 311/312 FIRE	NO	NO	NO	NO	NO	NO	NO
SARA 311/312 PRESSURE	NO	NO	NO	NO	NO	NO	NO
SARA 311/312 REACTIVITY	NO	NO	NO	NO	NO	NO	NO
EPA EXTREMELY HAZARDOUS SUBSTANCE	NO	NO	NO	NO	NO	NO	NO
CALIFORNIA SAFE DRINKING WATER ACT (Proposition 65)							
This product does not contain any chemical listed on the California Safe Drinking Water Act list (Proposition 65)							
US OSHA PROGRAMS							
PEL	YES	NO	NO	NO	NO	NO	NO
PSM	NO	NO	NO	NO	NO	NO	NO
CHEMICAL SECURITY PROGRAMS							
DHS CFATS	NO	NO	NO	NO	NO	NO	NO
CHEMICAL WEAPONS CONVENTION							
	NO	NO	NO	NO	NO	NO	NO
US DRUG ENFORCEMENT ADMINISTRATION							
DEA Controlled Substances	NO	NO	NO	NO	NO	NO	NO
CHEMICAL INVENTORY PROGRAMS							
DSL	YES	YES	YES	YES	NO	NO	NO
REACH Pre-registered List	YES	YES	YES	YES	YES	YES	YES
TSCA	YES	YES	YES	YES	YES	YES	YES
TSCA Reset Rule	All ingredients in this product comply with the U.S. EPA TSCA Inventory Notification Requirements Rule (40 CFR 710 Subpart B.)						
European Inventory of Existing Commercial Chemical Substances (EINECS)	YES	YES	YES	YES	YES	YES	YES
EU No-Longer Polymers List (NLP)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Philippines	YES	YES	YES	YES	YES	YES	YES
Japan	YES	YES	YES	YES	YES	YES	YES

Australia	YES	YES	YES	YES	YES	YES	YES
Korea	YES	YES	YES	YES	YES	YES	YES
China	YES	YES	YES	YES	YES	YES	YES
New Zealand Inventory of Chemicals	YES	YES	YES	YES	YES	YES	YES

16. OTHER INFORMATION

16.1	Original Preparation	1 August 2019: 17 Nov 2019, updated for sale
16.2	Revision History	
16.3	Prepared by	ADVANCED CHEMICAL SAFETY, Inc. PO Box 152329 San Diego, CA 92195 (619) 990-4908
16.4	Date of Printing	February 14, 2020

DEFINITIONS OF TERMS

16.5	A large number of abbreviations and acronyms appear on a MSDS. Some of these which are commonly used include the following:	
	Section 2	GHS: Global Harmonization System OSHA: U.S. Occupational Safety and Health Administration. CLP: Classification and Packaging WHMIS: Workplace Hazardous Materials Information System STOT: Specific Target Organ Toxicity
	Section 3	CAS #: Chemical Abstract Service index number EINECS #: European Chemical Substances Information System index number
	Section 5	NFPA: Nation Fire Protection Association Health Hazard: 0 (material that on exposure under fire conditions would offer no hazard beyond that of ordinary combustible materials); 1 (materials that on exposure under fire conditions could cause irritation or minor residual injury); 2 (materials that on intense or continued exposure under fire conditions could cause temporary incapacitation or possible residual injury); 3 (materials that can on short exposure could cause serious temporary or residual injury); 4 (materials that under very short exposure could cause death or major residual injury). Flammability Hazard Reactivity Hazard: Refer to definitions for "Hazardous Materials Identification System". Flash Point: Minimum temperature at which a liquid gives off sufficient vapors to form an ignitable mixture with air. Autoignition Temperature: The minimum temperature required to initiate combustion in air with no other source of ignition. LEL: The lowest percent of vapor in air, by volume, that will explode or ignite in the presence of an ignition source. UEL: The highest percent of vapor in air, by volume, that will explode or ignite in the presence of an ignition source.
	Section 8	ACGIH - American Conference of Governmental Industrial Hygienists, a professional association which establishes exposure limits. TLV - Threshold Limit Value - an airborne concentration of a substance which represents conditions under which it is generally believed that nearly all workers may be repeatedly exposed without adverse effect. The duration must be considered, including the 8-hour Time Weighted Average (TWA), the 15-minute Short Term Exposure Limit, and the instantaneous Ceiling Level (C). Skin absorption effects must also be considered PEL - Permissible Exposure Limit - This exposure value means exactly the same as a TLV, except that it is enforceable by OSHA. The OSHA Permissible Exposure Limits are based in the 1989 PELs and the June, 1993 Air Contaminants Rule (<i>Federal Register</i> : 58: 35338-35351 and 58: 40191). Both the current PELs and the vacated PELs are indicated. The phrase, "Vacated 1989 PEL," is placed next to the PEL which was vacated by Court Order. IDLH - Immediately Dangerous to Life and Health - This level represents a concentration from which one can escape within 30-minutes without suffering escape-preventing or permanent injury. The DFG - MAK is the Republic of Germany's Maximum Exposure Level, similar to the U.S. PEL. NIOSH is the National Institute of Occupational Safety and Health, which is the research arm of the U.S. Occupational Safety and Health Administration (OSHA). NIOSH issues exposure guidelines called Recommended Exposure Levels (RELs) . When no exposure guidelines are established, an entry of NE (Not Established) is made for reference.
	Section 11	LD₅₀ : Lethal Dose (solids & liquids) which kills 50% of the exposed animals; LC₅₀ : Lethal Concentration (gases) which kills 50% of the exposed animals; ppm: Concentration expressed in parts of material per million parts of air or water; mg/m³ : Concentration expressed in weight of substance per volume of air; mg/kg: Quantity of material, by weight, administered to a test subject, based on their body weight in kg IARC - the International Agency for Research on Cancer; NTP - the National Toxicology Program, RTECS - the Registry of Toxic Effects of Chemical Substances, OSHA and CAL/OSHA. IARC and NTP rate chemicals on a scale of decreasing potential to cause human cancer with rankings from 1 to 4. Subrankings (2A, 2B, etc.) are also used. TDLo, the lowest dose to cause a symptom and TCLo the lowest concentration to cause a symptom; TDo, LDLo, and LDo, or TC, TCo, LCLo, and LCo, the lowest dose (or concentration) to cause lethal or toxic effects. BEI - Biological Exposure Indices, represent the levels of determinants which are most likely to be observed in specimens collected from a healthy worker who has been exposed to chemicals to the same extent as a worker with inhalation exposure to the TLV.
	Section 12	LC₅₀: The lowest concentration in water which kills 50% of the test subjects. EC₅₀: The Effect Concentration in water at which 50% of the test species is affected.
	Section 13	US EPA Hazardous Waste Codes: refer to 40 CFR 261.20
	Section 14	DOT: US Department of Transportation IATA: International Air Transport Association IMO: International Maritime Organization MARPOL: International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978 IBC Code : Merchant Shipping Code
	Section 15	RCRA: US Resource Conservation and Recovery Act SARA: US Superfund Amendments and Reauthorization Act PSM: US OSHA Process Safety Management CFATS: US Department of Homeland Security Chemical Facility Anti-terrorism Standard DSL: Canadian Domestic Substances List NDSL: Canadian Non-Domestic Substances List REACH: European Registration, Evaluation, Authorization and Restriction of Chemicals list TSCA: US Toxic Substances Control Act